

K1439 — the “seven” saga CLOSES: the blind computation gives the fermion weight = $5/2$, not $7/2$ (Lyra retracts F947’s $7/2$). Casey asks “why are we stressing over 7?” — honest answer: two legitimate reasons, now both resolved, and it’s time to stop. ★ (1) It began as a real corpus-wide mislabel (“ $g=7$ is the Bergman genus,” ~347 places) that was corrupting the projector work — that HAD to be fixed (Hua $\rightarrow$ the scalar exponent is $5 = n_C$; K1429). (2) Then Lyra had banked the fermion weight as $7/2$ since F947 and built the exact projector on it — so we had to check it blind before trusting it. The blind check just came back: $5/2$ ($= n_C/\text{rank} =$ the scalar weight, no spinor shift). So the “7-as-fermion-weight” question is resolved *negatively*, the error is caught *before it shipped*, and we can close it. The 7 has a real home — it’s the signature ($5+2 =$ space+time directions), the vector dimension of $\text{so}(7)$ — it was never the fermion weight. ★ Side rulings: DROP the $35/4$ gap as any well-posedness justification (Elie MEASURED the actual gap $= \sqrt{2}|p|$, contains no g ; $35/4 = n_C \cdot g/4$ contains $g=7$ and doesn’t describe the operator — a circularity risk against the blind weight). ★ Grace FIRED the DESI discriminator on the K1436 data: the phantom crossing is ABSENT in the radial BAO — BST’s monotone relaxation survives the cleanest probe (honest scope: radial-BAO-alone; full combo still $\sim 3\sigma$, shared with ΛCDM). Cal cold-reads.

Keeper (2026-08-13, ~05:00. Casey asked why we’re stressing over 7 in the same round Lyra’s blind computation answered it — it’s $5/2$. The discipline worked: it corrected a weeks-old human assertion. And I’ll own that the auditor filed more notes on this one integer than it merited. It’s closed now. Cal cold-reads. Nothing pushed.)

★ ANSWER to Casey: why we stressed over 7, and why we stop now

Two legitimate reasons drove it, and both are resolved: 1. A real error was propagating. The corpus called $g=7$ “the Bergman genus” in ~347 places (K1429). That mislabel was quietly corrupting the whole projector construction (it’s what put the wrong $7/2$ exponent into Paper118 and into F947). Fixing it was not optional — it was a genuine bug, now fixed against Hua’s primary source: the scalar Bergman exponent is $5 = n_C$, and $5/6/7$ are three *distinct* $\text{so}(7)$ invariants (dual-Coxeter / Coxeter / signature), not one thing with three names. 2. Lyra had banked a wrong number and built on it. Since F947 she’d asserted the fermion weight = $7/2$ and constructed the exact projector on it. A banked, load-bearing number *must* be checked blind before it ships. So Cal, Grace, and I held the line — four rejections, compute it on the correct ($K^{1/2}$ -twisted spin) bundle, blind to the answer. Lyra did, and the geometry returned $5/2$, not $7/2$. She retracted it clean.

So the honest verdict on the “stress”: it was the immune system doing its job. A *seductive* near-miss — $g=7$ is a BST integer, “ $5+2=7=\text{signature}$ ” is pretty, and more than one of us (Lyra in F947, me in K1428/K1430) kept being pulled to make it the fermion weight. The discipline is exactly what caught it. The cost of the stress was the cost of a pretty wrong idea; the machine handled it. And I’ll own my part: the auditor filed ~15 notes on this one integer — more than it merited — partly because it kept generating corrections (several mine). Right-sizing going forward.

Where the 7 actually lives (so we’re placing it, not abandoning it): $g = 7$ is the signature — the vector dimension of $\text{so}(7)$, the $5+2$ of space-plus-time directions, and the exponent in the spinor *size* $2^{n_C}=32$. It’s a real, load-bearing BST integer. It simply is not the fermion’s Bergman weight — that’s $5/2 = n_C/\text{rank}$, the same as the

scalar. The fork (6 vs 7) resolved to NEITHER: the spinor rides at the base, no shift.

❓ Stop stressing over 7. The question is closed (negatively, cleanly). It is *not* load-bearing for the shippable deliverables: the QM paper doesn't need it; Grace's cosmology result doesn't need it; B1's credential needs the weight to be *right* (now 5/2), not to be 7. Refocus on the momentum: the paper at your desk, the sea one integral from the real domain, the cosmology discriminator just fired favorably.

★ Ratify — Lyra's blind correction (fermion weight = 5/2)

Lyra computed on the correct object (true spinor, $K^{1/2}$ twist), blind, and got $5/2 = n_C/\text{rank} =$ the scalar Bergman weight — no spinor shift. F947's 7/2 is retracted (she'll fix the earlier note). Cheap confirmation owed: Elie's $m \rightarrow 0 \rightarrow$ F947 check (the massless sea must collapse to the plain Bergman space; if the weight is 5/2, that's the object it returns to). This is the discipline working exactly as designed — the blind computation corrected the human's weeks-old assertion. Credit Lyra for the clean walk-back. Register the weight at 5/2, blind-derived (pending the $m \rightarrow 0$ confirmation).

★ The 35/4 gap — DROP it as justification (side with Elie's measurement)

Lyra says the gap is $35/4 = n_C \cdot g/4$ “derived from curvature.” Elie's 5216 measurement contradicts it: the built operator has $D^2 = 2|p|^2 \cdot I$ identically and spectral gap $\sqrt{2}|p|$ (zero at $p=0$), containing no g at all. So: - The operator's actual gap is $\sqrt{2}|p|$ + the mass term (Elie), not 35/4. Well-posedness holds for almost every p ; the mass term restores the uniform gap (Elie's “the gap is the mass” — confirmed on the real operator). - $35/4 = n_C \cdot g/4$ contains $g=7$ — the very number the blind weight computation must be blind to. Justifying well-posedness with a g -containing formula, then running the (blind) weight on that construction, is an upstream circularity (Elie's flag, Cal's “wood not marble”). DROP 35/4 as the well-posedness justification — it costs nothing and removes the circularity. If 35/4 is a genuine Lichnerowicz curvature eigenvalue it can be shown separately, computed blind; but the *built* operator's gap is $\sqrt{2}|p|$, and Elie's $D^2=2|p|^2 \cdot I$ shows no curvature term is currently in the operator — which itself flags whether the build is fully carrying the domain curvature (Lyra/Elie reconcile). Do not bank 35/4.

★ Ratify — Grace's cosmology discriminator FIRED (favorable, honest scope)

Grace fired the radial- D_H discriminator on the K1436 DESI DR2 table (verified at source, block-diagonal covariance, six independent D_H points — *not* the circular CPL fit): - The phantom-crossing fingerprint is ABSENT. The signature (monotone residual, negative low- $z \rightarrow$ positive high- z , crossing near $z \approx 0.9$) is *not* there; the residual pattern is $--++--$ (a mid- z bump at 1.3σ), no monotone trend, correlation ≈ 0 . The far (high- z) points are *negative* — the opposite of the phantom signature. So the phantom crossing DESI headlines is NOT in the most direct measurement of $H(z)$; BST's monotone relaxation-from-above survives the radial probe. - Honest scope (Grace kept it, correctly): this is radial BAO alone. The full combination (SNe + CMB) still leans dynamical/phantom at $\sim 3\sigma$, driven by the *supernovae*, and BST shares Λ CDM's tension there. So: on the cleanest direct probe BST is *not* falsified and the phantom is absent; on the full combo it's a live sub- 5σ tension BST shares with Λ CDM. - This is a genuine, honest, favorable result — and the “physics is on GitHub” payoff in action (Keeper pulled the data K1436, Grace fired the armed discriminator). Register at radial-BAO-alone tier; the discriminator that was armed-and-unfired for four days has fired. This is the kind of result worth more than the seven ever was.

Route

1. Lyra — fix F947 (weight $7/2 \rightarrow 5/2$); finish the two-point integral (the last mile); reconcile with Elie whether the built operator carries the domain curvature ($D^2=2|p|^2$ suggests kinetic-only) or is still effectively flat.
2. Elie — run $m \rightarrow 0 \rightarrow$ F947 (confirms the 5/2 weight + the transport); then the four census tests when the two-point integral lands.
3. Grace — cosmology result registered (radial-BAO phantom-absent, full-combo tension shared); consider whether the radial-vs-SNe split is itself a BST-distinctive statement (the phantom lives in the SNe, not $H(z)$) — a lead, honestly scoped. Then #76 remainder.

4. Cal — cold-read armed (4 checks); the weight is now $5/2$ not 7, so the “reproducing forces the weight” check confirms $5/2$ (not “zero at $7/2$ ”); drop the $35/4$ dependency.
5. Keeper — closed the seven saga (weight = $5/2$, blind); dropped $35/4$ as justification; ratified Grace’s cosmology fire. Right-sizing attention off the resolved integer, back onto the deliverables (QM paper GO-ready; cosmology; the sea’s last integral).

— Keeper, K1439, 2026-08-13. THE SEVEN SAGA CLOSES: blind computation (Lyra, on the correct $K^{1/2}$ -twisted spin bundle) gives fermion weight = $5/2 = n_C/\text{rank}$ = scalar weight (NO spinor shift), NOT $7/2$; F947’s $7/2$ RETRACTED; fork resolved to NEITHER 6 nor 7 (spinor at base). Casey “why stress over 7”: (1) real corpus mislabel “ $g=7$ =Bergman genus” ~347 places corrupting projector → fixed (Hua, scalar $\exp=5=n_C$; $5/6/7$ = dual-Coxeter/Coxeter/signature distinct); (2) Lyra banked $7/2$ since F947 + built on it → HAD to check blind → came back $5/2$. Immune system worked; caught a seductive near-miss ($g=7$ pretty, $5+2=7$ =signature); Keeper owns filing ~15 notes on one integer, right-sizing. 7’s real home = signature (vector dim $\text{so}(7)$, space+time $5+2$, spinor size $2^5=32$), NOT fermion weight. STOP stressing — not load-bearing for shippables (QM paper, cosmology); B1 needs weight RIGHT ($5/2$) not 7. RATIFY Lyra $5/2$ blind (pending Elie $m \rightarrow 0 \rightarrow$ F947 confirm); clean walk-back. DROP $35/4$ gap: Elie 5216 measured actual gap = $\sqrt{2|p|}$ ($D^2=2|p|^2 \cdot I$, zero at $p=0$, no g); $35/4=n_C \cdot g/4$ contains $g=7$ (circularity vs blind weight) + doesn’t describe operator → drop as justification (costs nothing); gap is the mass ($\sqrt{2|p|}$ +mass term); $D^2=2|p|^2$ flags whether build carries domain curvature (Lyra/Elie reconcile); don’t bank $35/4$. RATIFY Grace cosmology: fired discriminator on K1436 data (verified source, block-diag cov, 6 indep D_H pts, not CPL) → phantom crossing ABSENT in radial BAO (pattern $--++--$, mid- z bump 1.3σ , no monotone trend, high- z NEGATIVE = opposite of phantom); BST monotone relaxation survives cleanest probe; HONEST SCOPE radial-BAO-alone (full combo SNe+CMB $\sim 3\sigma$ dynamical, driven by SNe, BST shares Λ CDM tension); genuine favorable result, physics-on-GitHub payoff (Keeper K1436 pull → Grace fire). Route: Lyra fix F947 $7/2 \rightarrow 5/2$ + two-point integral + reconcile curvature ($D^2=2|p|^2$ kinetic?); Elie $m \rightarrow 0 \rightarrow$ F947 + 4 census tests; Grace register cosmology + radial-vs-SNe-split lead + #76; Cal cold-read (weight now $5/2$, drop $35/4$); Keeper closed saga, dropped $35/4$, ratified Grace, right-sizing. Cal cold-reads. Nothing pushed.